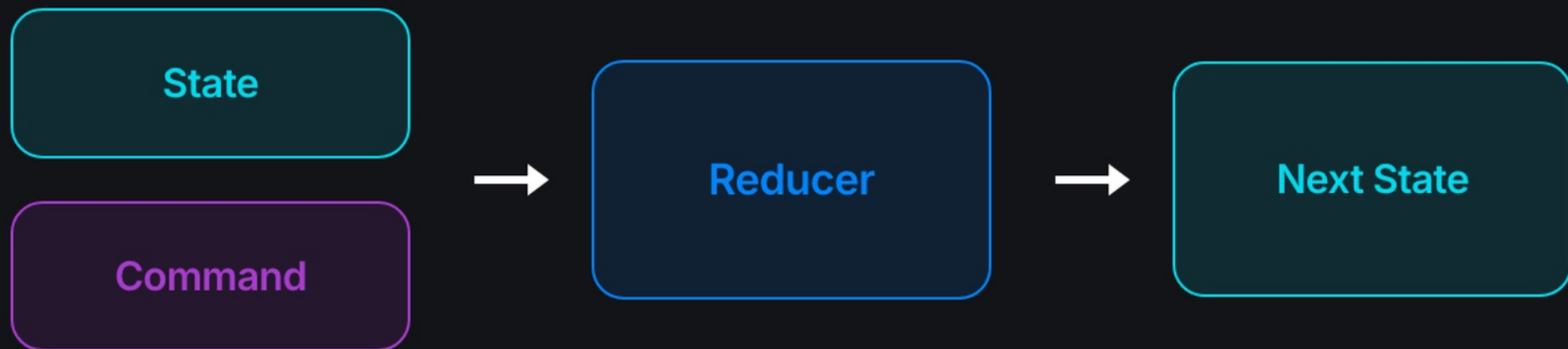


The Reactor

Local-first documents that sync.

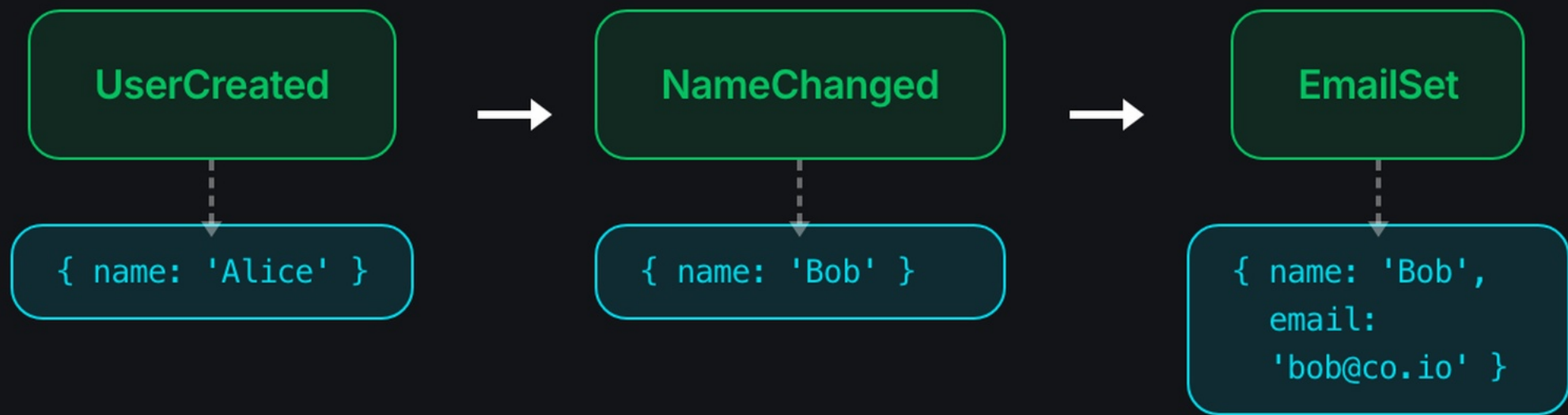
Reducer Update Loop

Unidirectional state transitions via pure functions



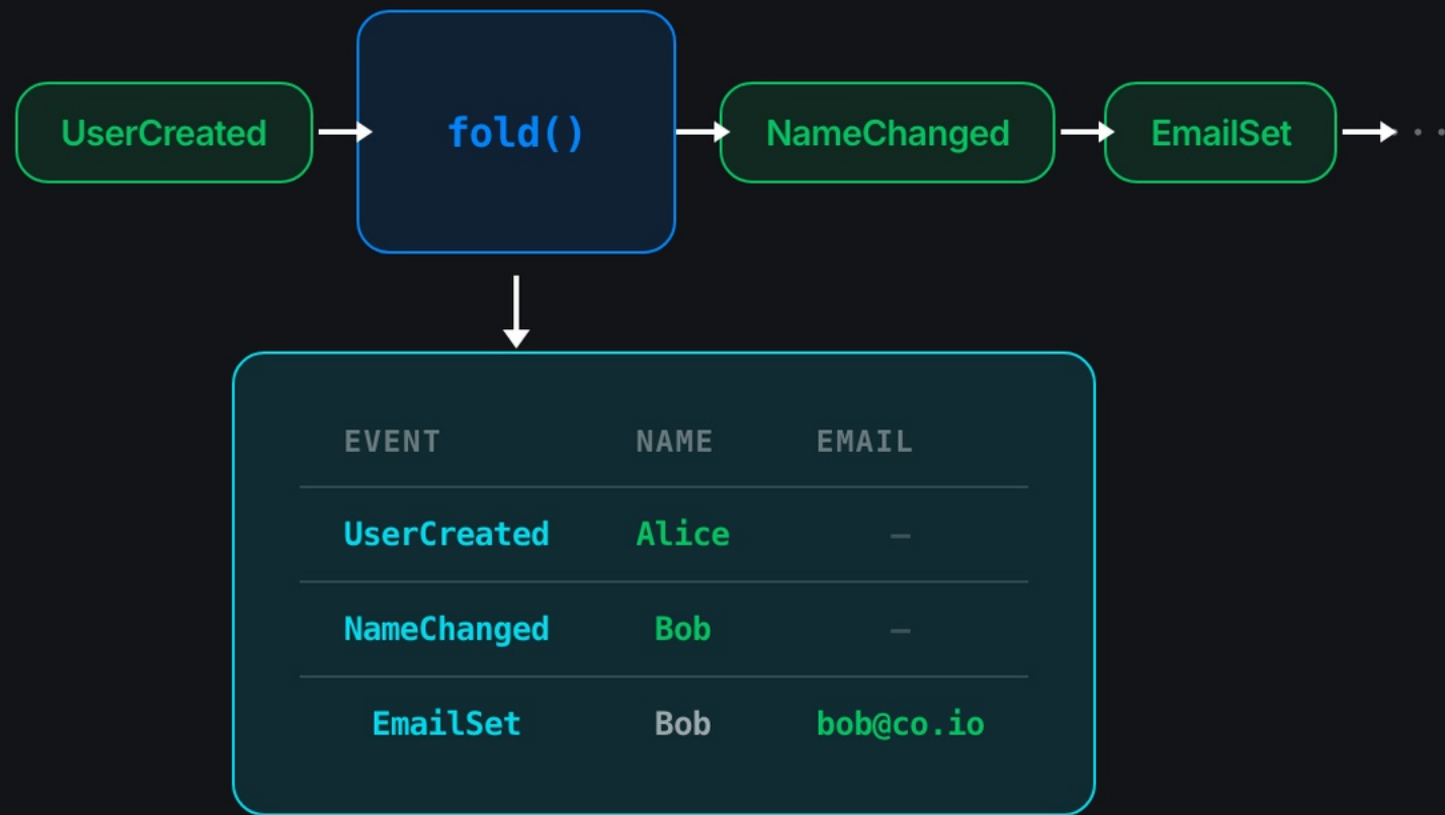
Event Sourcing

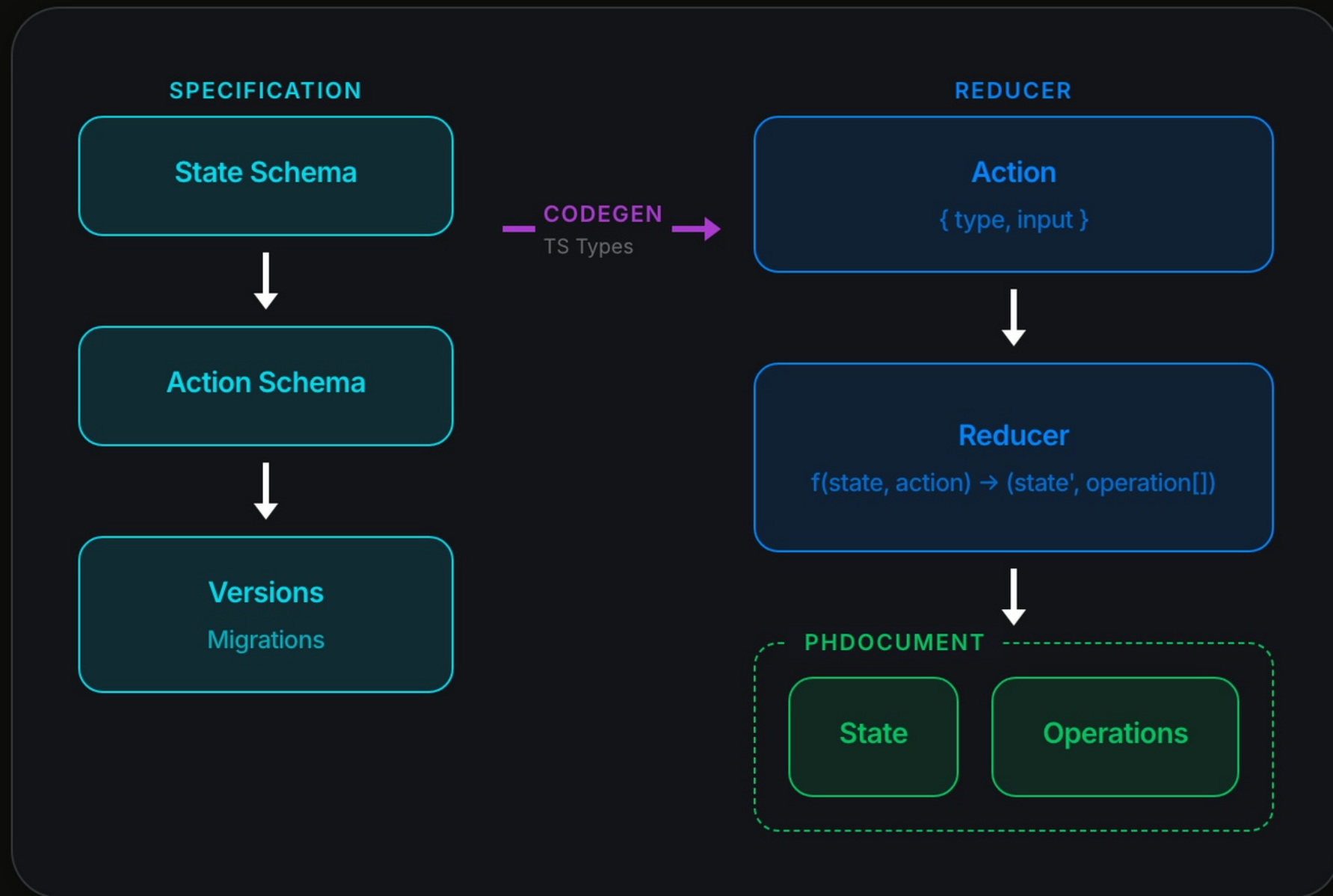
State as a stream of events



Aggregate

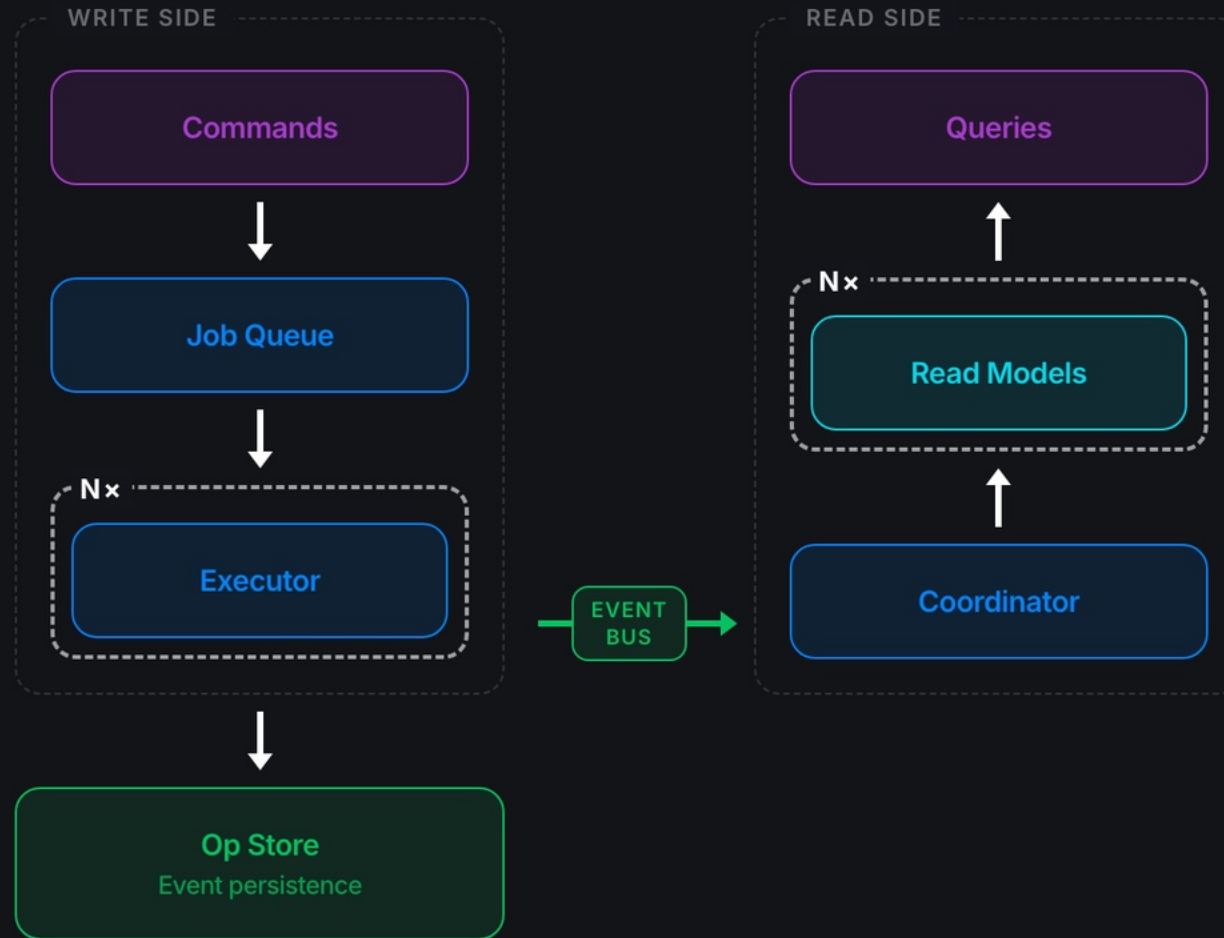
Folding a stream of events into a result

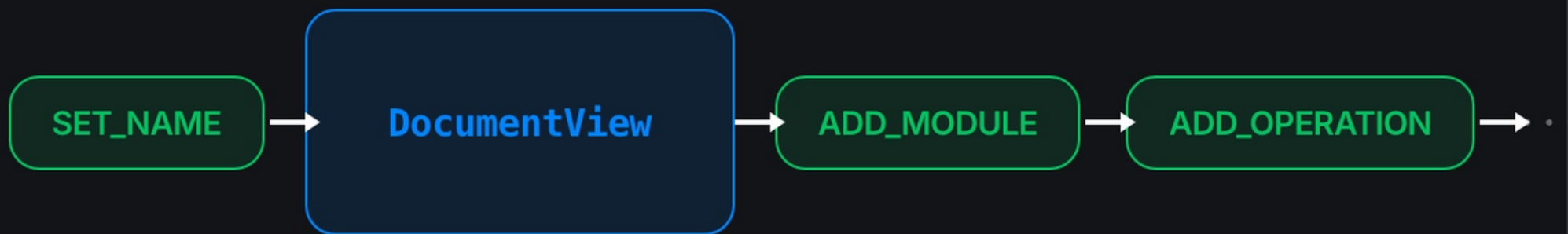




Reactor CQRS

Write/read side separation with independent scaling



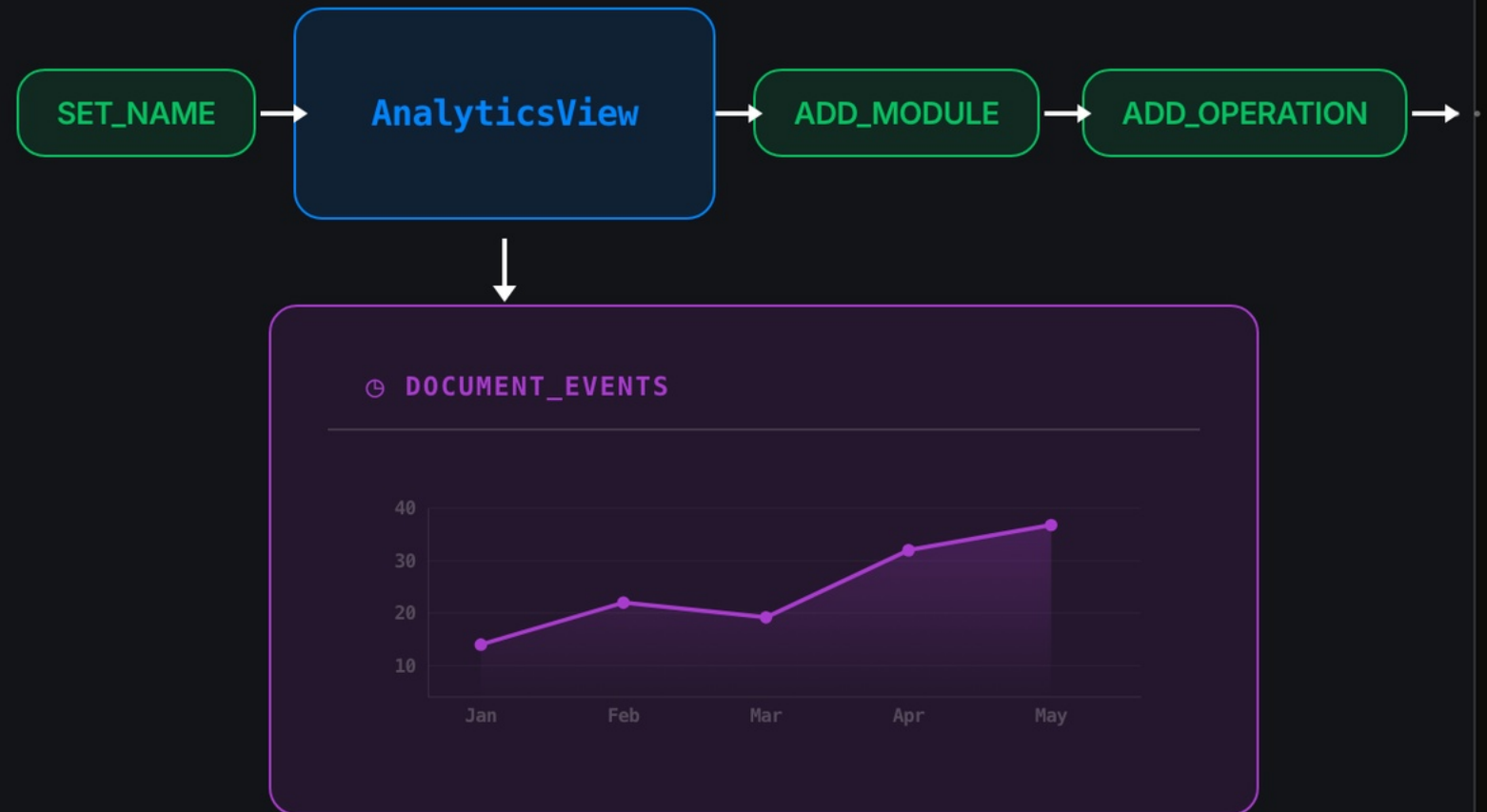


▼ DOCUMENTS

ID	STATE	MODEL	UPDATED_AT
doc_a1	{ "name": "Budget Q1", ... }	budget/v1	2025-01-15
doc_b2	{ "total": 4200, ... }	invoice/v2	2025-02-03

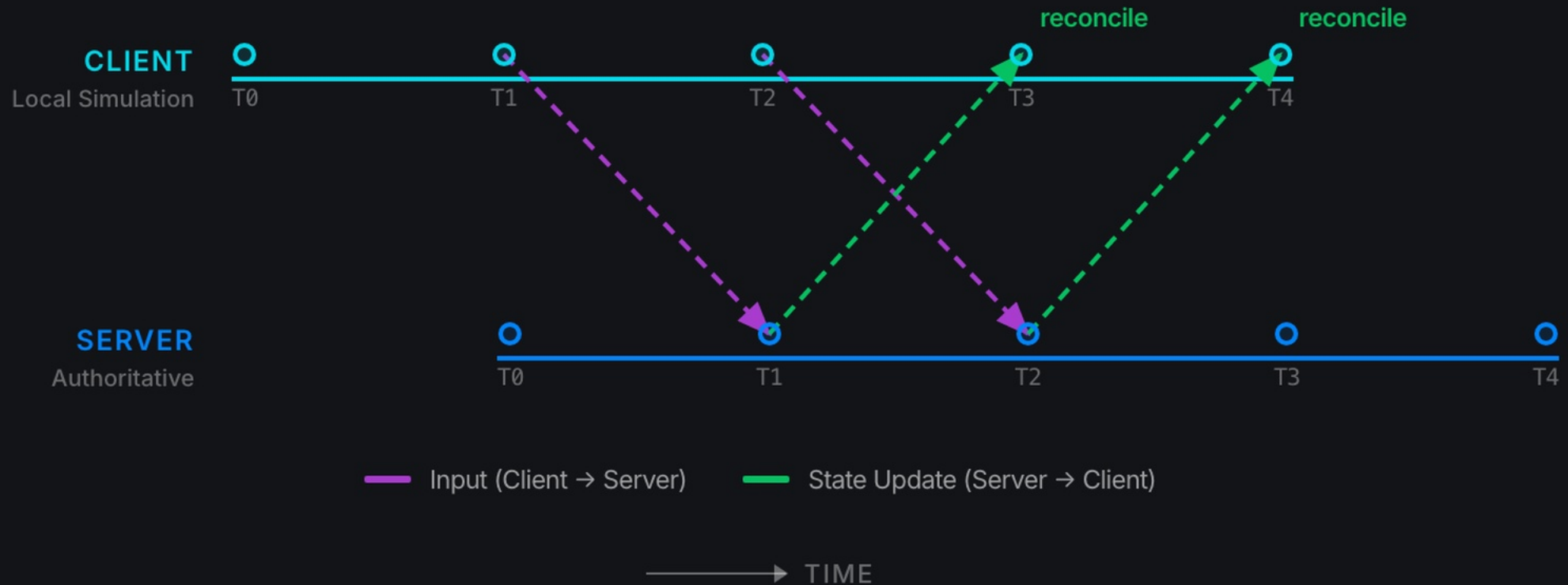
Analytics Read Model

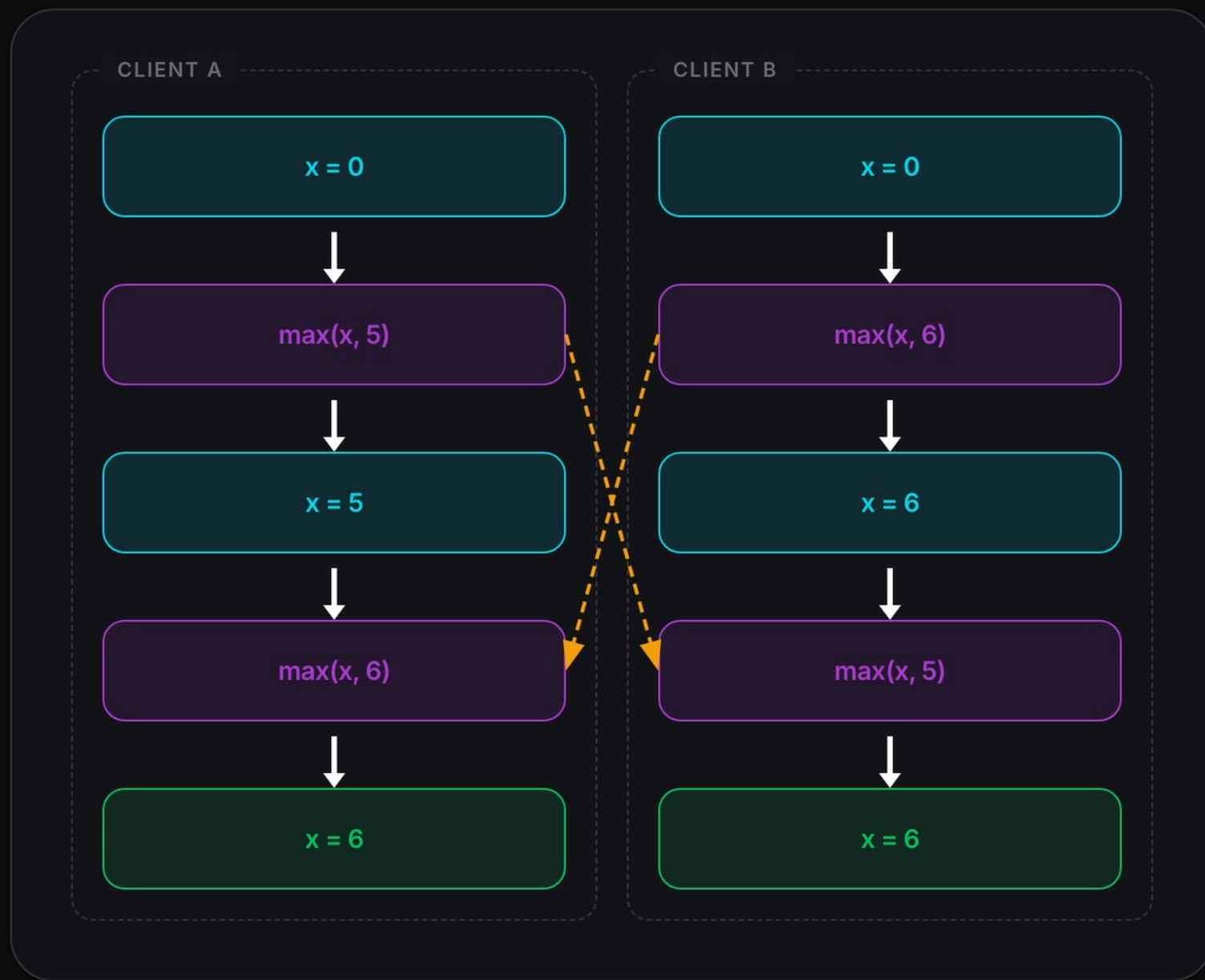
Collecting change analytics into a time-series database

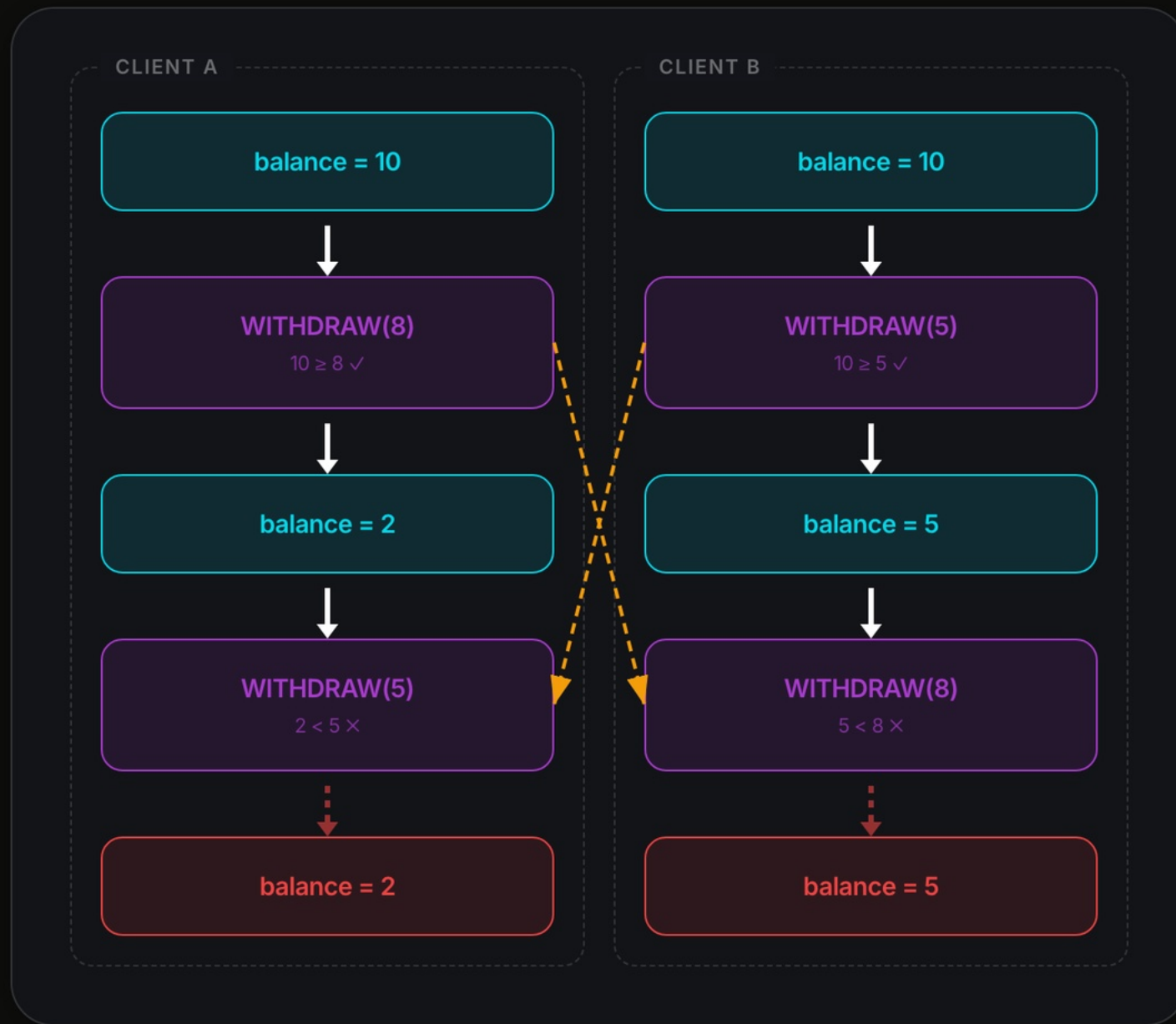


Client / Server Game Timelines

Local simulation runs ahead; authoritative simulation confirms







Interleaving Breaks State Hashes

Each operation records its previous state hash — interleaving invalidates the chain



Every operation after the first mismatch must be rewritten with new hashes.

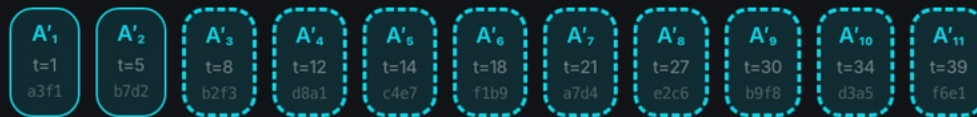
Reactor A

Reactor B

reshuffled ops

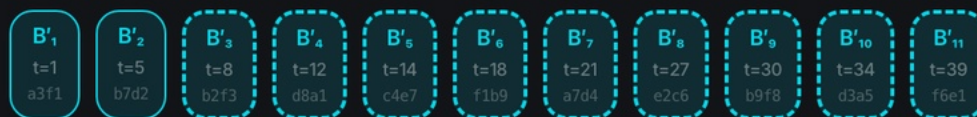
reshuffled ops

GARBAGE COLLECT



A' — 11 OPERATIONS

=



B' — 11 OPERATIONS

A' = B' — identical streams

Payload-only signature

DEFAULT

PREVIOUS OPERATION



NEXT OPERATION



✗ **Independent of state**
signature remains valid even when state changes due to reshuffle

Payload + input-state hash

HIGH SECURITY

PREVIOUS OPERATION

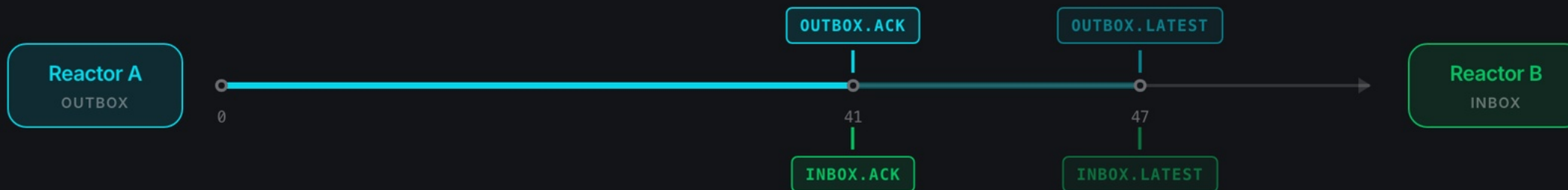


NEXT OPERATION

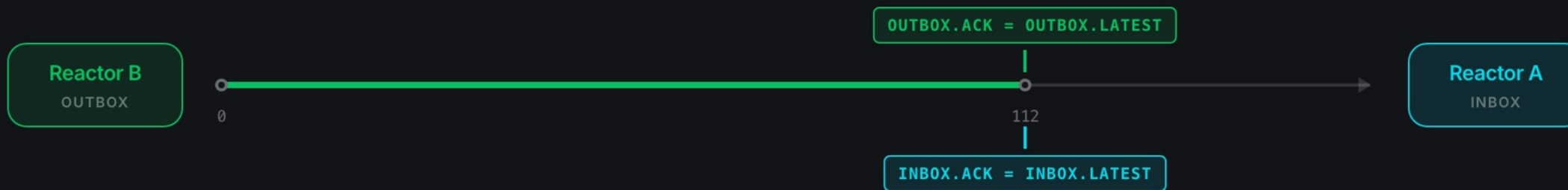


✓ **Bound to state**
action requires an explicit state

A → B · A'S OUTBOX / B'S INBOX · A'S ORDINALS



B → A · B'S OUTBOX / A'S INBOX · B'S ORDINALS



ack ≤ latest

cursors only move forward

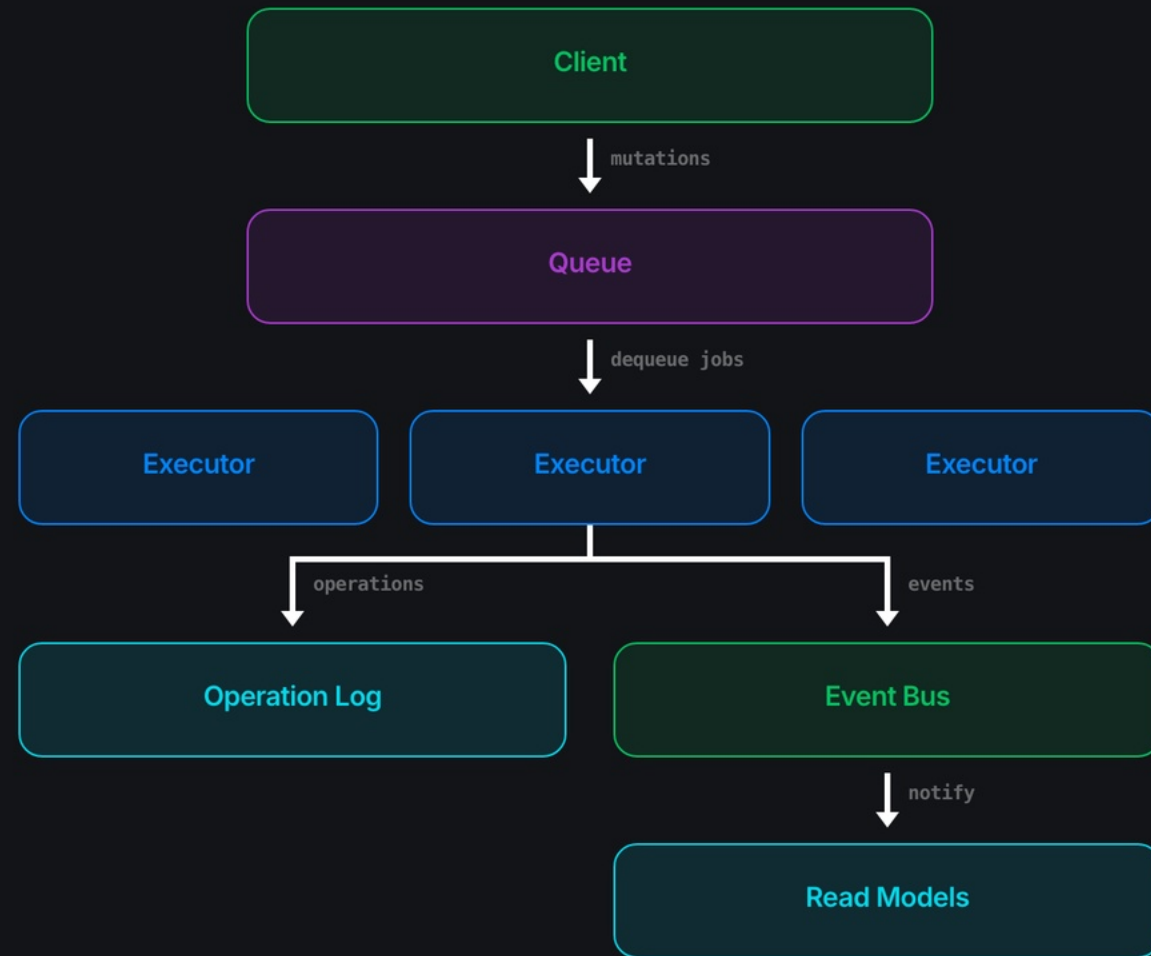
synced ⇔ ack == latest

Thanks

Questions?

Reactor Architecture

Job processing pipeline



Payload + input-state hash + id

CRITICAL SECURITY

PREVIOUS OPERATION

id

state

+

action

NEXT OPERATION

id

state

+

```
{
  type: "SIGN_STATEMENT",
  payload: { ... },
  signature: sig(payload + H(state) + id)
}
```



Bound to state and order

action requires specific state and order

